Multivariate Analysis of Scotch Whisky Components

To Cover: - A high-level description of the study and its motivation. - A quantitative descriptive analysis of the data. - A graphical/qualitative analysis of the data. - Can you assume that the data follow a multivariate normal distribution? - no - Are there "surprising" observations? You must try to answer the question (with a proper discussion): Is the cluster analysis consistent with different distances and linkages?

Abstract

Introduction

High level description of the study and its motivation

Seeing if it's possible to classify whiskey as counterfeit using trace element analysis from total reflection x-ray fluorescence (TXRF). The study also attempted cold ashing, but causes contamination - and they deemed it is unnecessary

As an alternative to ICP How good was it - pretty good it seems, can correctly discriminate between non-counterfeit and counterfeit observations with a linear discriminator - TXRF should be easier to carry out compared to ICP

Techniques used in the study are: - PCA - denodograms - linear discrimnator

Data

The data contains contains 32 observations across: - Counterfeit - Blend - Grain - Highland - Lowland - Speyside

With trace elements for, (P, S, Cl, K, Ca, Mn, Fe, Cu, Zn, Br and Rb), measured in mg/L.

12 out of the 32 observations have at least one of the trace element detected below LOD. All the counterfeit whiskey observations also have at least one variables below LOD

Amount of observations below LOD by individual element:

P	S	Cl	K	Ca	Mn	Fe	Cu	Zn	Br	Rb
10	0	1	0	0	0	0	0	0	1	3

Transformations on Data

Handling the LOD values

If we were to drop observations with trace elements that were measured below LOD, we'd lose more than a third of the dataset, including all counterfeit observations, because of this they will instead be substitued.

The two common ways of substituting LOD values are, $LOD/2$ and $LOD/\sqrt{2}$. Similar to the paper that this data is from, this report will be subtituing all LOD values with $LOD/2$.

Table 1: Table continues below

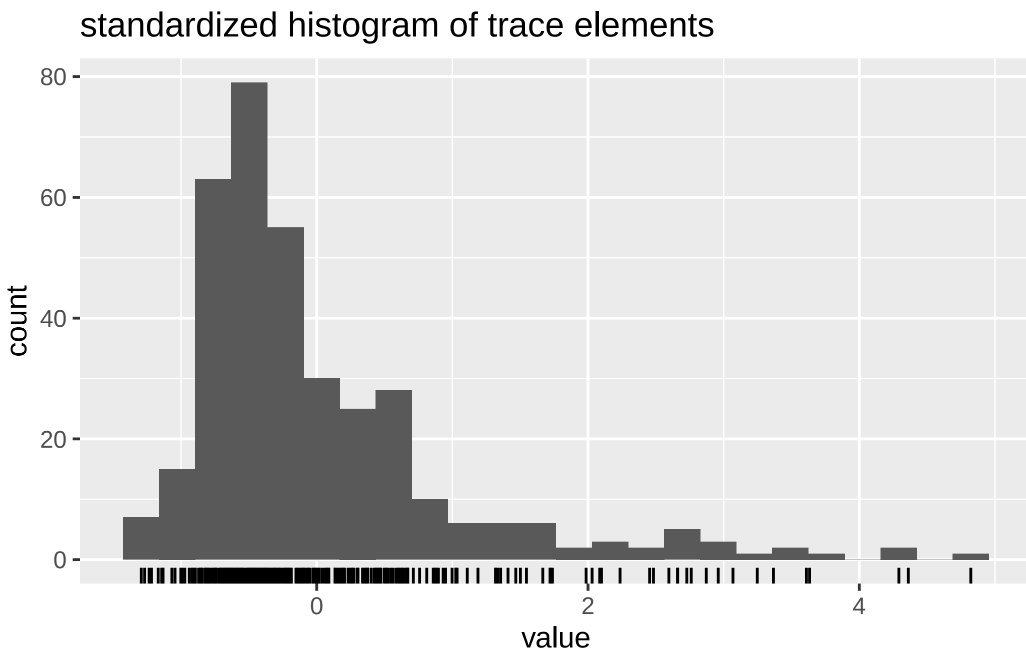
	S	Cl	K	Ca	Mn	Fe	Cu
Min.	0.576	0.0415	0.336	0.416	0.006	0.019	0.038
1st Qu.	1.08	0.1427	3.11	0.7432	0.01175	0.04925	0.183
Median	2.585	0.188	6.165	1.045	0.0195	0.088	0.2405
Mean	5.019	0.2691	10.26	1.192	0.02322	0.09456	0.3799
3rd Qu.	5.54	0.3433	12.8	1.4	0.03125	0.129	0.5255
Max.	26.1	1.35	37.7	4.13	0.053	0.288	1.32

	Zn	Br	Rb
Min.	0.007	0.001	5e-04
1st Qu.	0.01875	0.003	0.002
Median	0.023	0.004	0.006
Mean	0.03716	0.007781	0.00925
3rd Qu.	0.03725	0.007	0.01325
Max.	0.194	0.068	0.039

From the summary we can see all values are above 0, but they aren't all in the same magnitured of range. E.g., Br is a lot smaller than P.

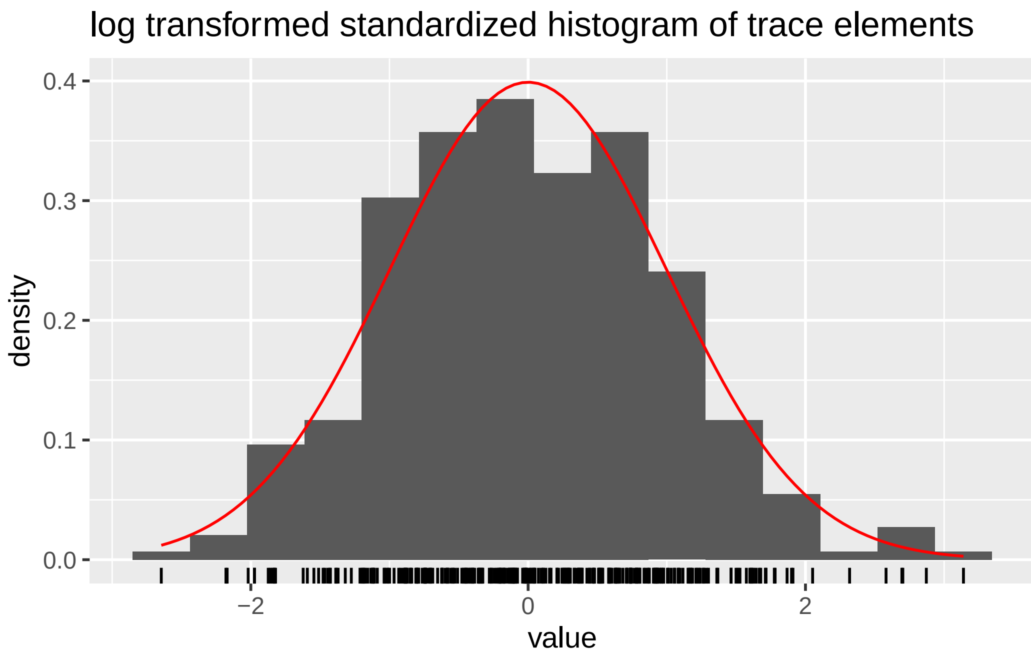
Log Transform on the data

Lastly, looking at (standardized) histogram of all the variables, we can see it looks rather skew,



After applying a log transform, the data looks a lot closer to a normal distribution.

```
Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
i Please use `after_stat(density)` instead.
```

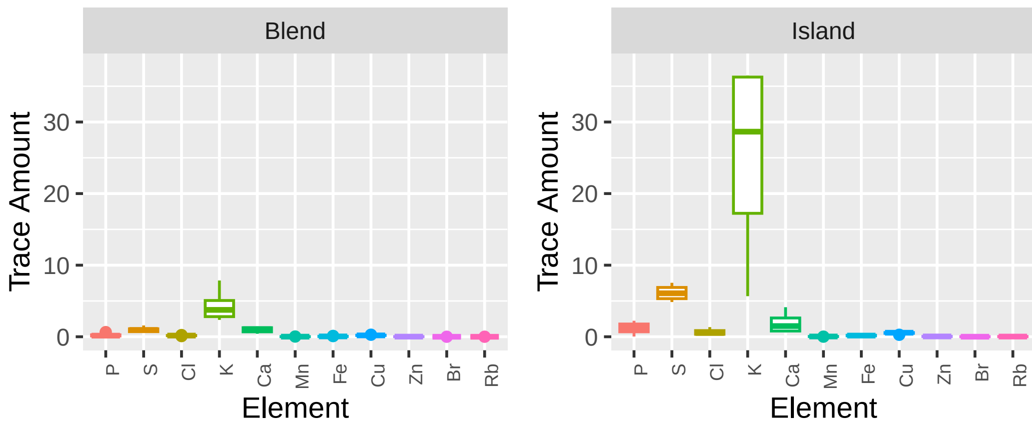


A Qualitative Analysis of Data

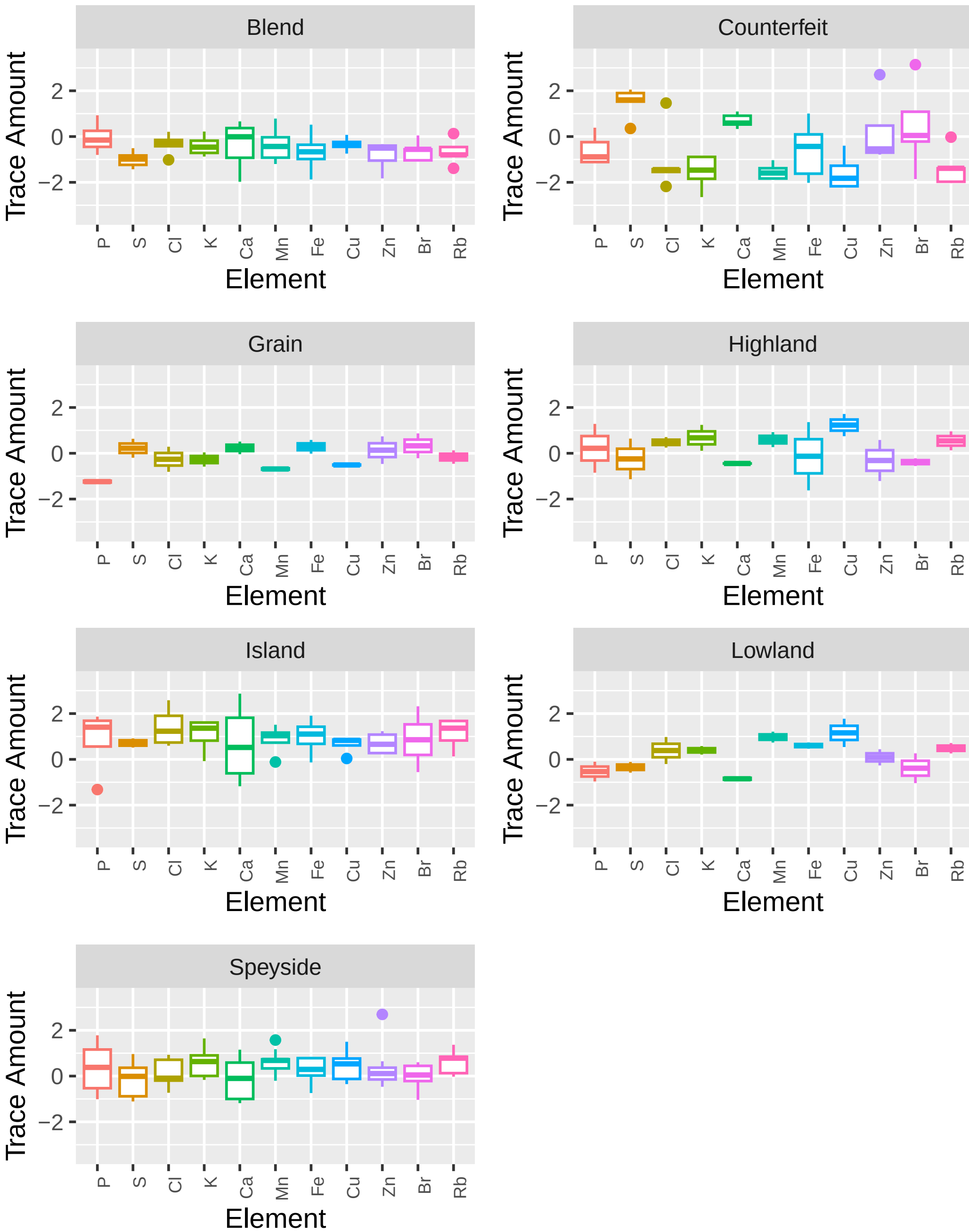
Box Plots

First looking at box plots of elements across Descriptor: Because K's magnitude is so great compared to the other trace elements, standerdizing the data helps better compare across all trace elements.

You can see an example below of why it's important,

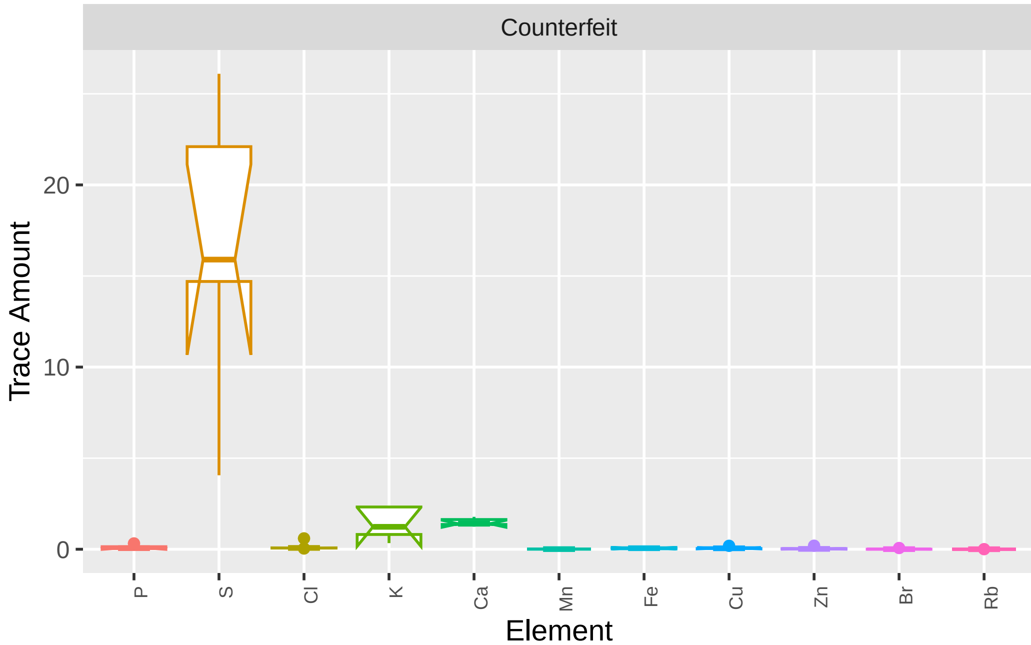


After standardization it's easier to compare. Looking at the boxplots there seem to be some differences. Counterfeit stands out a lot because of the high amount of **S** it has compared to the others. Lowland has a interestingly little amount of variance across all of it's elements.

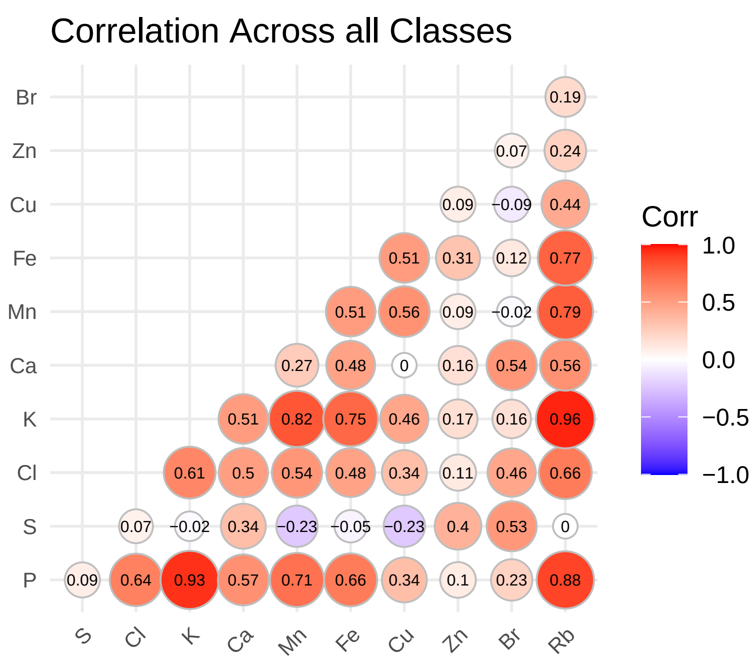


Note on Notches for box plots

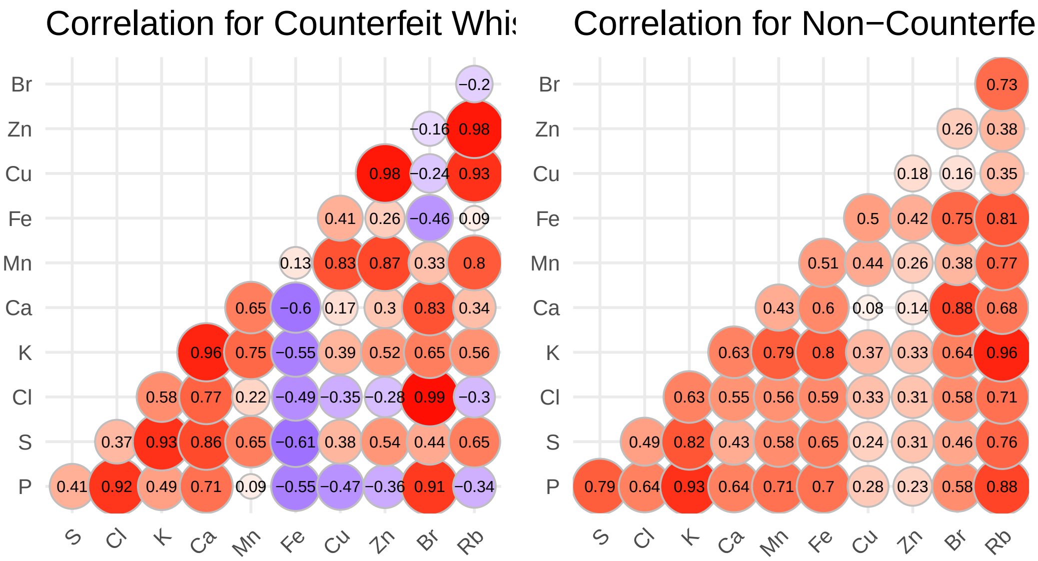
As the sample size is tiny, any attempt at adding notches causes it to go outside the range of the hinges (outside of the IQR). For an example:



Correlograms



K, **P**, and **RB** all have strong postiive correlations with one another,



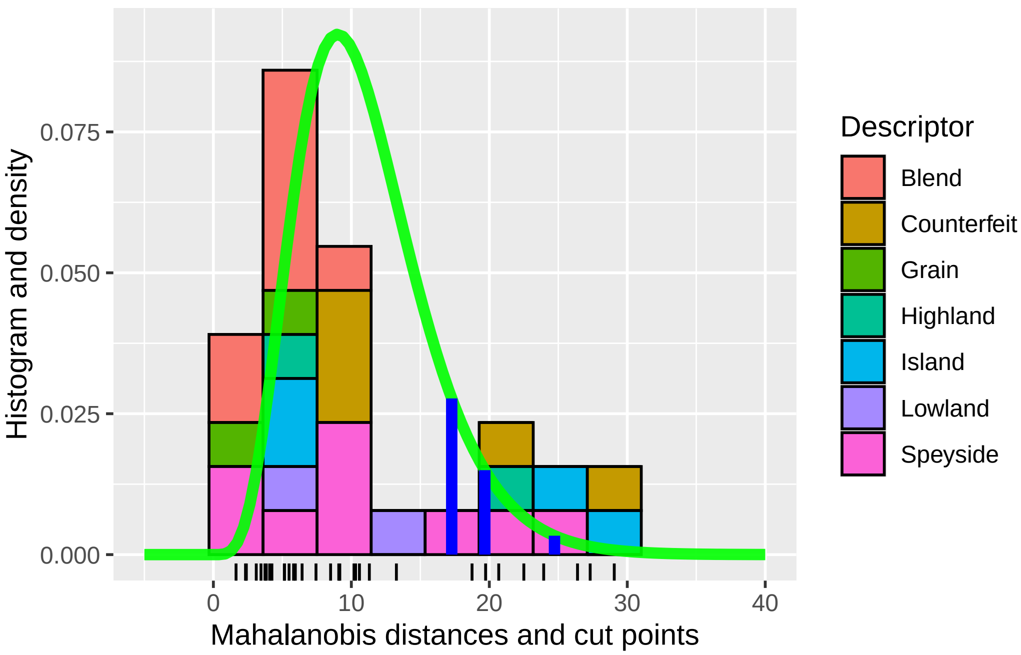
There's a clear indicator for some difference with a lot more negative correlations for the counterfeit whiskeys compared to the others. Particularly along **Fe**. Although, the strongest correlations are still those positive ones.

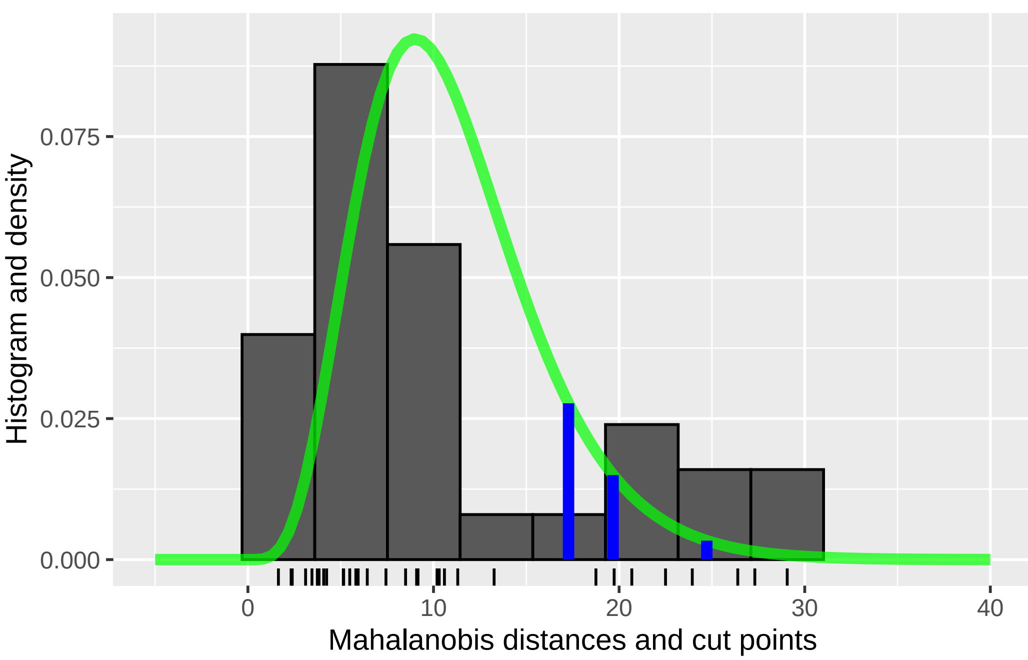
When separating out based on Counterfeit vs Non-Counterfeit, Non-Counterfeit still matains those strong correlations, with the greatest still being that between, **K**, **P**, and **RB**. This might be a common trace element picked up during whiskey production.

Due to the unbalanced observations, it makes sense that the original correlogram is similar to the non-counterfeit correlogram.

It's worthwhile also checking across more descriptor labels

Mahalanobis:

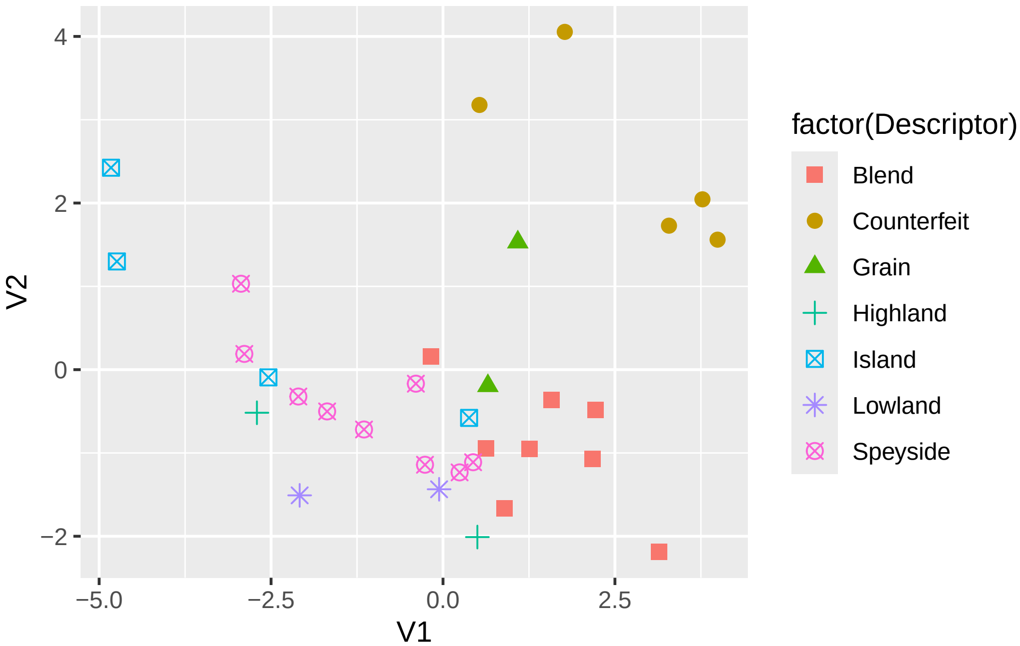




The data seems to be forming 2 groups, with a dip in the middle (around 15 on the x axis). The only thing that stands out (for what might be the reason) is observations with 'Blend' as their descriptor are all to the left, dragging everything else with them.

Its important to note the histograms with descriptor colours isn't scaled 100%, ngl, I have no idea how to do it correctly, but for a qualitative analysis it's close enough.

TODO: see what those points on the far right are ### PCA



There is a clear separation of the counterfeits from the rest in the upper right corner. Grain seems to be seeping into that cluster a bit tho.

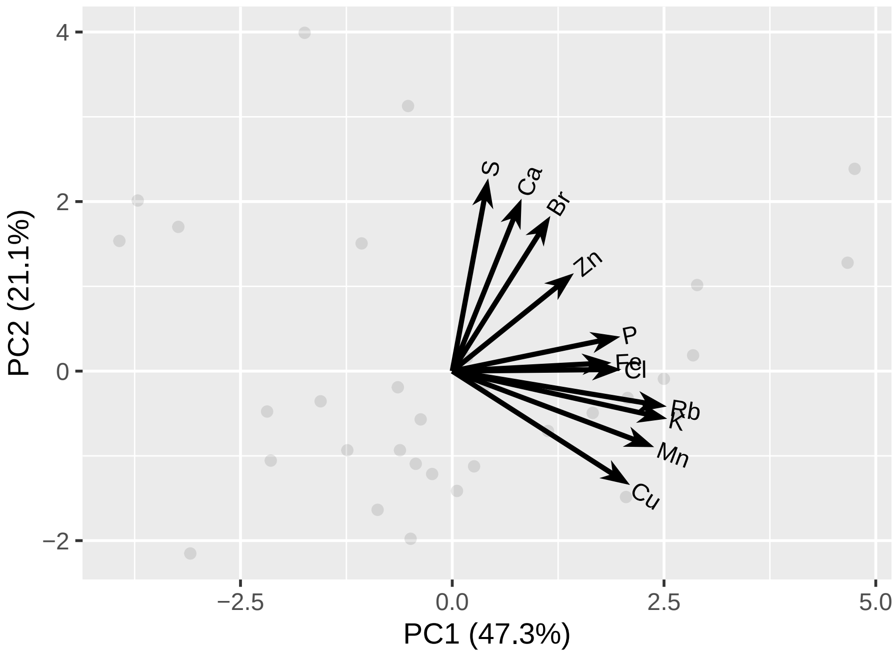
```
pca2 <- prcomp(whiskey.stand[-(1:2)], center=TRUE, scale=TRUE)
summary(pca2)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4     PC5     PC6     PC7
Standard deviation     2.2801 1.5242 1.0951 0.86707 0.66526 0.60480 0.49676
Proportion of Variance 0.4726 0.2112 0.1090 0.06835 0.04023 0.03325 0.02243
Cumulative Proportion  0.4726 0.6838 0.7928 0.86119 0.90142 0.93468 0.95711
                          PC8     PC9    PC10    PC11
Standard deviation     0.4337 0.41989 0.30011 0.13164
Proportion of Variance 0.0171 0.01603 0.00819 0.00158
Cumulative Proportion  0.9742 0.99024 0.99842 1.00000
```

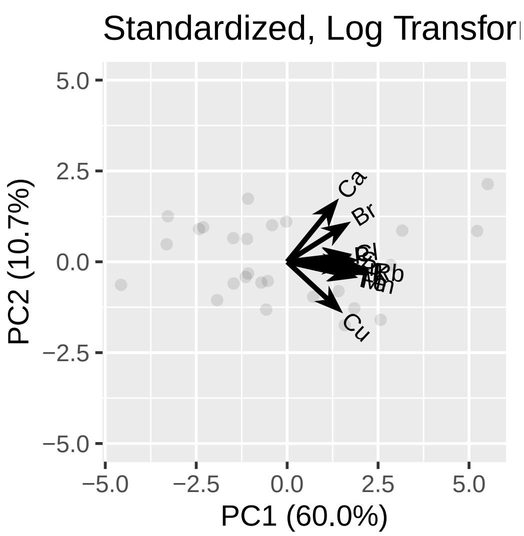
```
pca2$rotation
```

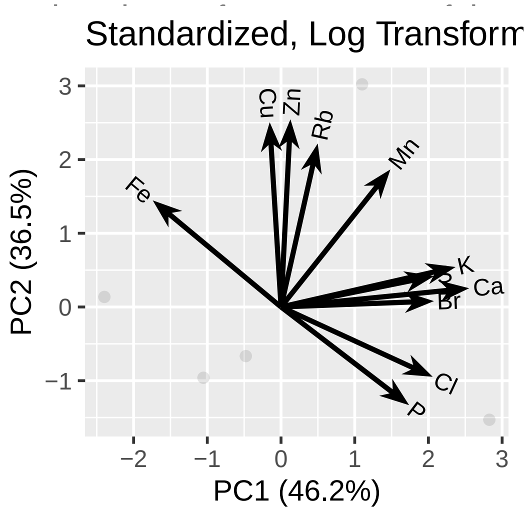
```
         PC1          PC2        PC3          PC4        PC5         PC6
P  0.31924612  0.097902279 -0.2120991  0.548308225 -0.4074035 -0.25314862
S  0.06821158  0.547348073  0.2311134  0.210395591  0.1663372 -0.55567168
Cl 0.32116572  0.004685254 -0.4160660 -0.448679450 -0.0934019 -0.16342892
K  0.40817466 -0.135223577 -0.1112254  0.108624481  0.1853779 -0.19881070
Ca 0.13165642  0.489406943 -0.2908339  0.276487520  0.1177837  0.69073588
Mn 0.38346175 -0.215726408 -0.1068212  0.075797703  0.2868130  0.08295689
Fe 0.30232008  0.023820062  0.4770248 -0.001570168 -0.6435641  0.23811392
Cu 0.33758824 -0.323300917  0.1261756 -0.121665818 -0.1012310  0.04957821
Zn 0.23091370  0.277915821  0.5483864 -0.262856115  0.2841972  0.09431934
Br 0.18633290  0.440497018 -0.2534662 -0.514524933 -0.1860483 -0.05855964
Rb 0.40730144 -0.100474472  0.1080633  0.105614098  0.3568282  0.07369217
          PC7          PC8         PC9         PC10         PC11
P  -0.52869122 -0.083990891  0.01374212  0.052852472  0.152499874
S   0.42569948 -0.064619956 -0.23955641  0.137134474  0.007039065
Cl  0.17855867 -0.657997370  0.08626514 -0.056947504  0.106981736
K   0.06810047  0.186195148  0.20370771 -0.405506709 -0.686882950
Ca  0.12212179 -0.134296674 -0.18704658 -0.058824485 -0.140472616
Mn  0.11459330  0.129053270  0.20840316  0.791746388 -0.006246692
Fe  0.37278420  0.009221927  0.26356842  0.007676463 -0.032042842
Cu -0.02656161  0.032501876 -0.85355614  0.031125182 -0.082421925
Zn -0.55131742 -0.278623606  0.09711992  0.048835360 -0.128675339
Br -0.13914520  0.614347047 -0.01634251  0.043246938  0.072781350
Rb  0.10822873  0.173940948  0.08568159 -0.418469216  0.666329348
```

```r
ggbiplot(pca2, obs.scale = 1, var.scale = 1, alpha = .1)
```



TODO: Inteperut this more

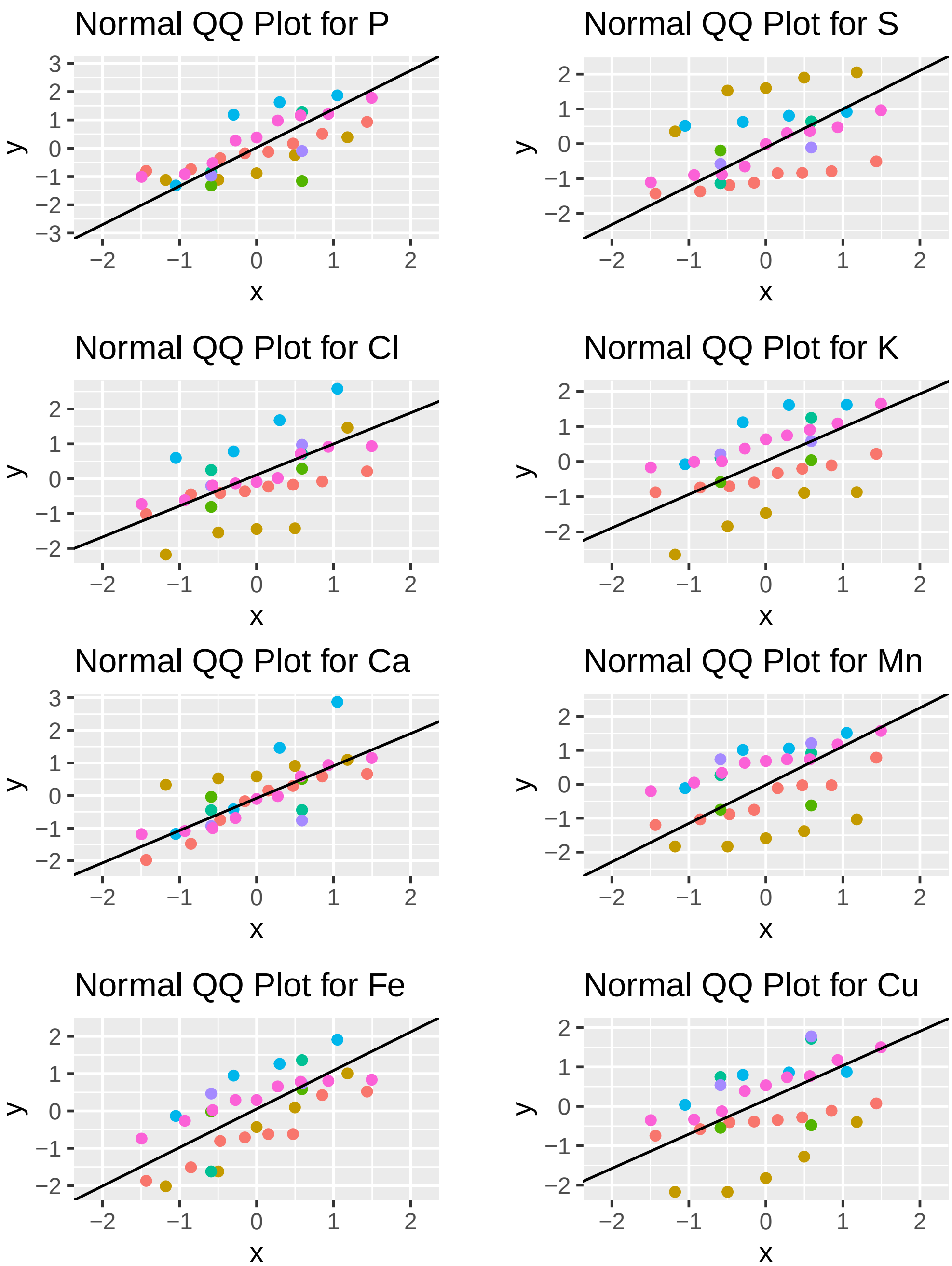


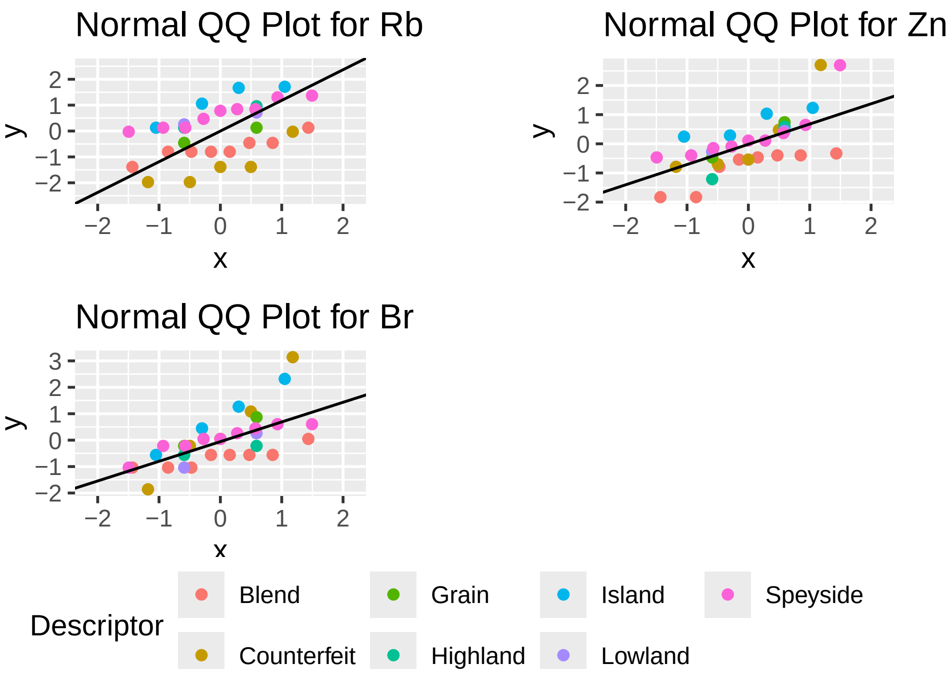


Once again, similar to the bar plots, immedietly the **Fe** axis jumps out, being negative across PC1.

Multivariate Normality

TODO: reconsider if the above histograms are needed


Normal QQ Plot for P
Normal QQ Plot for S
Normal QQ Plot for Cl
Normal QQ Plot for K
Normal QQ Plot for Ca
Normal QQ Plot for Mn
Normal QQ Plot for Fe
Normal QQ Plot for Cu
x
y



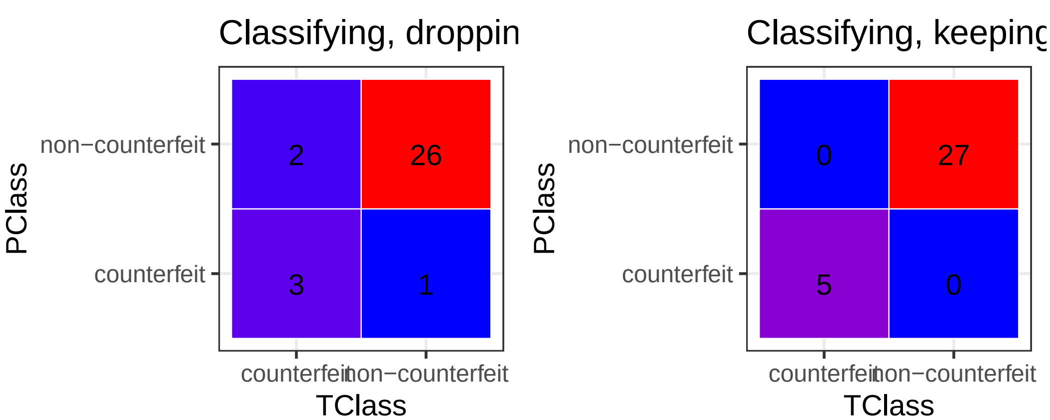
Interestingly you can make out the counterfeits seem to be the strange observations in **S**, and **Cu**.

Apart from that the Q-Q plots look a-okay

TODO: Anderson Darling Test - can't remember the exact name :p

Classifiying by distance

Building confusion matrix:



using mahalanobis distance causes a fair bit of error when dropping the ith row for counterfeit observations. Might be because of the poor amount of data?

distance has to be done using PCA, since there would be too many factors compared to the amount of counterfeit observations ($p > 5$).

TODO: try classifying across all Descriptors